ModelIt!

Student Activity Pack

Engineering Earthquake-Proof Skyscrapers

Modeling the Physics of Structural Survival Under Seismic Forces

Grade 10L3 | NGSS HS-ESS2-2, HS-PS2-1 | Structure and Function

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do some buildings collapse in earthquakes while others nearby survive with little damage?

2 What do you think is the difference between a rigid building and a flexible building during an earthquake?

3 Draw what you think happens to a tall building when seismic waves hit its foundation.

4 Have you heard of tuned mass dampers or base isolation? What do you think they do?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Seismic Wave Intensity | Building Mass | Damper Effectiveness | Foundation Flexibility | Resonance Frequency | Material Strength | Floor Acceleration | Evacuation Time | Damage Index

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Moderate Earthquake, Modern Design

0.3g PGA with advanced damping and base isolation

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Resonance Catastrophe

Seismic frequency matching building natural period

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Maximum Credible Earthquake

1.5g PGA near-fault ground motion

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Soil Amplification Factor — The degree to which local soil conditions amplify or dampen seismic waves before they reach the foundation — soft clay can amplify ground motion by 2-5 times compared to bedrock

Aftershock Vulnerability — The increased susceptibility to damage during aftershocks when the primary earthquake has already weakened structural connections and reduced the building's remaining capacity

Non-Structural Damage — Damage to interior elements — ceilings, partitions, mechanical systems, elevators — that can injure occupants and block evacuation even when the structural frame remains intact

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How can a million-ton building survive forces that level entire cities — and why do some skyscrapers collapse while others barely sway? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that resonance, not raw force, is the primary cause of catastrophic building failure in earthquakes?**

2. **Why is the three-layer defense (isolation, flexibility, damping) more effective than making the building as strong as possible?**

3. **What trade-offs does your model reveal between building mass, height, and seismic vulnerability?**

4. **How would your model change if the building were on soft clay versus bedrock?**

5. **What does 'earthquake-proof versus earthquake-resistant' mean in terms of your Damage Index results?**

8 STEM Challenge: Design a Seismic Protection System for a 60-Story Tower

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YOUR ENGINEERING MISSION

Design a comprehensive seismic protection system for a 60-story tower in a Zone 4 earthquake region that minimizes Damage Index while maintaining acceptable construction costs and Evacuation Time.

THE SCENARIO

A major city in an active seismic zone has approved construction of a new 60-story mixed-use tower. The site is 8 km from an active fault capable of producing M7.5 earthquakes. Your team must design the building's seismic protection system — including foundation type, structural system, damping technology, and evacuation plan — to achieve a Damage Index below 0.2 for the maximum credible earthquake.

Engineering Design Thinking

1. WHAT STRUCTURAL MATERIAL AND CONFIGURATION WILL YOU USE AND WHAT IS ITS STRENGTH-TO-WEIGHT RATIO?
2. WHERE WILL YOU PLACE DAMPERS IN THE BUILDING AND WHAT TYPE PROVIDES THE BEST MULTI-MODE ENERGY ABSORPTION?
3. HOW WILL YOU DESIGN EVACUATION ROUTES THAT REMAIN FUNCTIONAL EVEN WHEN THE BUILDING SUSTAINS DAMAGE?
4. WHAT IS THE ESTIMATED ADDITIONAL COST OF YOUR SEISMIC SYSTEM AS A PERCENTAGE OF TOTAL CONSTRUCTION COST?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Structural Engineers specializing in seismic design develop earthquake-resistant buildings, bridges, and infrastructure. They use computational modeling to simulate seismic response and work at engineering firms, government agencies, and research institutions, earning $80,000-$160,000/year. Seismologists who study earthquake physics earn $70,000-$130,000/year.